Assignment – 1.12

Introduction to Networking and the OSI Model Theory

Q - 1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

- **Wireshark Capture – Protocols Seen:**

  I installed Wireshark on my laptop, started a capture on my main network interface (Wi-Fi), and visited instagram.com. After stopping the capture, I noticed the following protocol names in the packet list:

- **Protocols Seen:**

  DNS
  TCP
  TLS
  HTTP
  QUIC
  UDP
  ARP
  ICMP

Q - 2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

➢ Packets and Their OSI Layers:

No.	Protocol	Info (Example)	OSI Layer	Reason
1	HTTP	GET request to instagram.com	Application Layer (Layer 7)	HTTP is used to request web pages and data.
2	TCP	SYN, ACK	Transport Layer (Layer 4)	TCP provides reliable data delivery between devices.
3	TLS	Client Hello	Presentation Layer (Layer 6)	TLS encrypts data to keep it secure.

Q - 3. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

➢ OSI Model with Real-World Examples:

Layer	Name	Function	Real-World Example
7	Application	Provides network services to applications.	WhatsApp messages
6	Presentation	Handles data format, encryption, and compression.	Instagram images and videos
5	Session	Manages connections and sessions.	Video calls
4	Transport	Provides reliable or unreliable data delivery.	TCP
3	Network	Handles routing and logical addressing.	IP
2	Data Link	Controls access to the physical media and detects errors.	Wi-Fi / Ethernet
1	Physical	Sends raw bits over the physical medium.	Wi-Fi radio signals

Q - 4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Hint: Think about how your message is typed, packaged, sent over the internet, and received.

- **How a WhatsApp Message Travels Through the OSI Model:**

1. **Application Layer:** We type and send a message on WhatsApp.

2. **Transport Layer:** It helps transfer the message from one device to another.

3. **Network Layer:** It finds the path for the message to reach the receiver.

4. **Data Link Layer:** It sends the data through Wi-Fi or a local network.

5. **Physical Layer:** The data travels through radio waves, cables, or other signals.

6. **Receiver:** The message reaches the receiver's phone and appears in WhatsApp.